Revision | **Source** Lecture review slide | **File** 10 / 11

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The review self-study exercises, fully solved

تمارين المراجعة محلولة بالكامل

These are the three exercises Dr. Al-Hadhrami closes the chapter with, set for self-study. They are solved here step by step.

الشرح بالعربي

هذه تمارين المراجعة الثلاثة في آخر ملف الدكتور. الفكرة المشتركة بينها: **ارسم مخطط فن واملأ المناطق الأربع بالأرقام** (A فقط، B فقط، المشترك، خارج الاثنين)، ثم يصبح كل فرع مجرد جمع لمناطق تراها أمامك. وتذكّر أن **neither ... nor** تعني دائمًا $1 - P(A \cup B)$.

EX 1 · BURGERS AND PIZZA

A study shows that about 37 percent of school children eat burger, 17 percent eat pizza, and 10 percent eat both burger and pizza. What is the probability that a randomly selected school child eats **neither** burger nor pizza?

1. **Extract:** P(B) = 0.37, P(Z) = 0.17, P(B ∩ Z) = 0.10.
2. **"Neither...nor" is the complement of the union.** First get the union — they overlap, so Rule 2:

   $$P(B \cup Z) = 0.37 + 0.17 - 0.10 = 0.44$$

3. $P(\text{neither}) = 1 - P(B \cup Z) = 1 - 0.44 =$ **0.56**

Cross-check with a Venn diagram: burger only = 0.37 − 0.10 = 0.27; pizza only = 0.17 − 0.10 = 0.07; both = 0.10; outside = 1 − (0.27 + 0.07 + 0.10) = 0.56. ✔

EX 2 · SMART PHONE COLORS

If the probabilities are, respectively, 0.23, 0.15, 0.21, and 0.09 that a person purchasing a new smart phone will choose the color gold, silver, black, or white, what is the probability that a given buyer will purchase a new phone that comes in one of those colors?

1. A phone has exactly one color, so the four events are **mutually exclusive** → extended Addition Rule 1.
2. $P(\text{gold} \cup \text{silver} \cup \text{black} \cup \text{white}) = 0.23 + 0.15 + 0.21 + 0.09 =$ **0.68**

Read what this tells you: the total is 0.68, not 1 — so 32% of buyers choose some other color entirely. A question that follows up with "find the probability the buyer chooses a different color" is answered by 1 − 0.68 = 0.32.

EX 3 · SET ALGEBRA WITH PROBABILITIES

If $P(A) = 0.3$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine: (a) $P(A^c)$ (b) $P(A \cup B)$ (c) $P(A^c \cap B)$ (d) $P(A \cap B^c)$ (e) $P((A \cup B)^c)$ (f) $P(A^c \cup B)$

Build the four-region picture first. A Venn diagram with two overlapping circles has exactly four regions, and every part of this question reads straight off it:

REGION	MEANING	PROBABILITY
A only	$A \cap B^c$	$0.3 - 0.1 = 0.2$
Both	$A \cap B$	0.1
B only	$A^c \cap B$	$0.2 - 0.1 = 0.1$
Neither	$(A \cup B)^c$	$1 - 0.4 = 0.6$
Total	S	**1.0** ✔

1. **(a)** $P(A^c) = 1 - P(A) = 1 - 0.3 =$ **0.7**
2. **(b)** $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.2 - 0.1 =$ **0.4**
3. **(c)** $P(A^c \cap B)$ = "B but not A" $= P(B) - P(A \cap B) = 0.2 - 0.1 =$ **0.1**
4. **(d)** $P(A \cap B^c)$ = "A but not B" $= P(A) - P(A \cap B) = 0.3 - 0.1 =$ **0.2**
5. **(e)** $P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.4 =$ **0.6**
6. **(f)** $P(A^c \cup B) = P(A^c) + P(B) - P(A^c \cap B) = 0.7 + 0.2 - 0.1 =$ **0.8**
   Or from the diagram: everything except "A only" $= 1 - 0.2 = 0.8$ ✔

THE TECHNIQUE THAT MAKES THIS QUESTION TRIVIAL

Never attack set-algebra questions with formulas alone. Draw the two circles, fill in the four regions as numbers, and every sub-part becomes an addition of regions you can see. Parts (c), (d) and (f) take longer to write than to solve once the diagram exists.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests
01 · Lesson 4-1 — Sample Spaces and Probability
02 · Lesson 4-2 — The Addition Rules
03 · Lesson 4-3 — Multiplication Rules and Conditional Probability
04 · Lesson 4-4 — Counting Rules
05 · Lesson 4-5 — Probability and Counting Rules
06 · Master formula sheet
07 · Which rule do I use?
08 · The 14 traps that cost marks
09 · Review exercises, solved ← this file
10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.